Investigating central factors for learning in digital workshops in art museums

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Abstract: This paper presents a study about processes supporting possibilities of student learning in digital workshops in art museums in Denmark. The investigation aims at providing insight into effects of selected factors important for learning in settings where digital media is used and dialogic communication is the approach. To address the research question of what conditional and processual factors can be argued to support learning potential, a model is developed theoretically and analyzed using structural equation model (SEM) on data collected in museums after the workshops (n=502). Results suggest that dialogic performance of museum educators, a positive art experience and qualities in the process of working with digital media are factors which can be argued to strongly support students' participation and reflection. The findings also show that in case students are disinterested and amotivational in meeting art and see little value in experiencing art works, their level of amotivation is likely to be decreased by these factors.

Keywords: museum education, art learning, digital media, young people, semi-formal, outcome, measurement

Introduction

The work of art museums in school service programs is an important activity for students' familiarity with, and learning about, art and issues presented in images and installations of art works. In these educational activities students are invited to discuss interpretations and meanings in exhibitions to learn about topics and issues relevant to society. In ARKEN Museum of Modern Art and The National Gallery Denmark, selected art works are the starting point for such endeavours, and efforts to support students' discussion and reflection are planned in three hour digital workshops, which are part of school and museum arrangements to visit art exhibitions. These efforts aim at engaging students in debating content and facilitating learning experiences involving a challenge to their views and attitudes towards art, as well as introducing them to ways of perceiving art as a resource for reflecting further in future occasions after the museum visit.

Evaluation of these activities and efforts are often performed within a qualitative approach, which provides insights into individual or group learning experiences and learning processes as conceived in the particular situation. While these types of evaluations provide rich knowledge of, for example, student interaction and discussion about art motifs among peers seen from the student's perspective,[1] or the advantages of using digital media,[2] the objective of the present study is to examine how selected factors influence learning possibilities in digital workshops from a general perspective. The central question is: from the viewpoint of art museums, are the efforts in digital workshops likely to facilitate learning?

The study establishes a conceptual model of the relationships between relevant dimensions of the workshop elements and student outcome, which are mediated by how motivated students feel and how they participate in discussion during activities. The model is based on preliminary fieldwork in the two museums, which aimed at clarifying important factors. The fieldwork resulted in an initial theoretical map of relations between students' perceptions of preconditions for learning, motivational aspects of the interaction, participation in dialogues and the student outcome. The operationalization draws on both the ideas explicated in the notion of latent phenomena, such as "perception of the dialogic performance of the museum educator", which are put forward in the theory of scale

[1] See how young people use different codes in the articulation of experiences and for interpreting art works in Helene Illeris's work (Illeris 2005).
[2] See how mobile blogging platforms link art museum and school activity in Palmyre Pierroux's and her colleagues article (Pierroux, Krange, & Sem 2011).

construction in measurement (DeVellis 2003, p.9), and on the view of constructs as socially constructed concepts, which take meaning in particular contexts through interaction and language use, as explicated in the concepts of “share-holding” and “co-authorship” in the work of Ragnar Rommetveit (2003, p.214). The constructs found important for learning processes are measured by a set of indicators (Kline 2010). Thus, the measurement of constructs and their relations are theoretically derived within a holistic approach which produces a map of relations contained in the model. This is then further developed empirically using techniques of exploratory and confirmatory factor analysis and structural model estimation, which support arguments for the effect of selected factors influencing learning within the aggregated generalized perspective.

Background and fieldwork

Digital workshops are educational events primarily aimed at upper secondary school and high school students, who visit the museums as part of a curriculum or a thematic project designed and planned by the teachers. The type of workshop included in this study uses still photo or audio-file recording as frame and tool to create small narratives of 2 to 3 minutes length edited in software packages such as Garage Band on Mac computers. Student learning experience in four types of workshops: “The wild 80s”, “Media used in art”, “National identity” and “Portrait, staging and identity” are surveyed in the study. These workshops conform to the general scheme of three elements: 1) students are guided on a tour in the galleries and relevant art works are selected for the purpose of discussing a theme or issue relating to the exhibition, 2) students are given a task to solve, which links to the theme and which involves their co-creation of a digital file in the form of recorded speech or a photo accompanied by text in group work, and 3) a presentation of the products in class, which entails a collective evaluation of the process of creating the file.

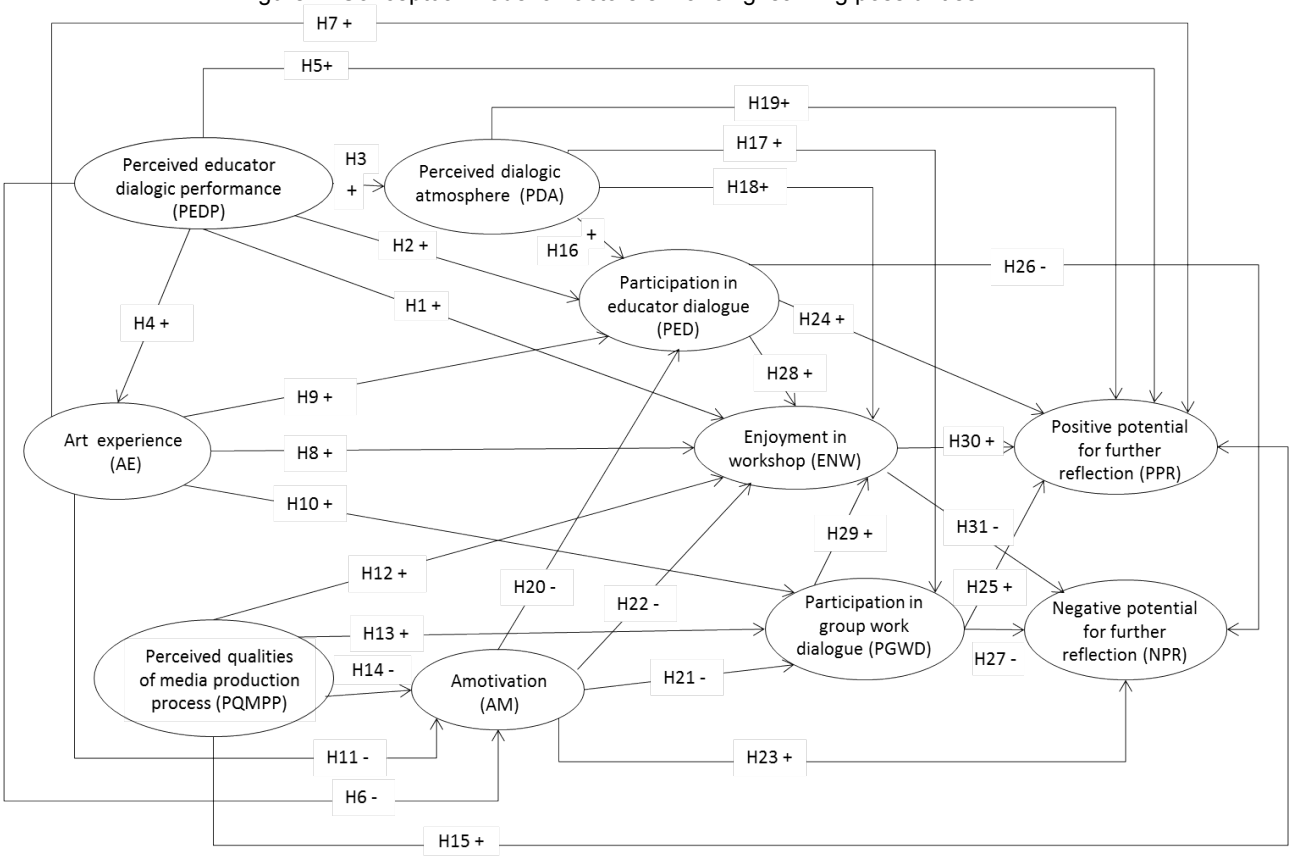
The concept of national identity is, for instance, debated through active co-production and negotiating of content and meanings in groups. Students are free to choose any material to create a scenario that they see illustrative of their view on the issue. Digital photos are then recorded and the group select the best and write short texts to illustrate its meaning. For example a group formed the Danish flag out of four red mattresses and placed a student on each of them to illustrate that although we as a population are different with varying lifestyles, affiliations and attitudes, the flag symbolize unification. Ideas and understandings of what it means to connect to a national identity were amply discussed within a setting of productivity.

The preliminary observations of work processes and student behavior provided a basis for conceptualizing central themes in this educational context. Motivational aspects related to student experience of the art works, the way the educator communicated content and invited students to give their opinions, and the process of actively producing seemed important dimensions of the workshops to understand relevant conditions for learning. The observations revealed that students participated with comments, and in some incidents also posed questions. But in some cases the students’ behavior also seemed rooted in disinterest and low motivation to engage, as some were chatting in the background despite the educator’s invitation to join in. Following these observations it was relevant to ask how a general pattern of factors enhancing learning possibilities would look like. Does the level of amotivation impact the opportunities to learn? On the basis of the preliminary observations and theoretical scrutiny a conceptual framework and a model were built to be tested against data.

Conceptual framework

With the initial observations as points of departure, a conceptual theoretical model was developed including a set of hypotheses to formally organize the analyses. See figure 2.

Figure 2: Conceptual model of factors enhancing learning possibilities



The model shows each construct as an oval with a name and abbreviation in parenthesis. In each oval a set of indicators are theorized to reflect the construct in question, they are then used as questions in the questionnaire. To reduce complexity in figure 2 the 53 indicators are not shown. Each arrow denotes an association assumed to link the constructs (either 'positive' or 'negative', where 'negative' means that an inverse relationship is assumed, e.g. art experience is assumed to reduce amotivation as in H11 -). The conceptual framework thus implies the idea of latent variables, which can be measured within a reflective measurement process using a set of indicators (Kline 2010, p.280). In the following section, constructs and their relations are presented and it is discussed how these can inform establishing the entire holistic model used to enquire into factors enhancing learning possibilities.

The three preconditional factors

The art educator plays a central role as initiator and organizer of events. Dysthe notes that dialogue and conversation are about qualities of the way participants take part in conversation. In education, students will tend to participate if the educator performs well in supporting these qualities (Dysthe 2003). Thus, the construct 'perceived educator dialogic performance' (PEDP) which she contends relates to the qualities of dialogue, i.e. conversation, as mutually symmetric and shared, acceptance of equal legitimacy of expressions, and the enabling of multi-voicedness (Dysthe 2002, p.342). In the context of art gallery education, particularly 'authentic questioning', the educators' use of student verbal contribution for further conversation is crucial. When the educator creates a welcoming atmosphere, invites all to participate, demonstrates interest in student opinions, and use responses purposefully, it is likely that this relates to a perceived dialogic atmosphere (PDA). This positive atmosphere supports participation in educator dialogue (PED), relates to the overall enjoyment of the workshop (ENW), and it will have an association with positive potential for further reflection (PPR). Moreover, it is assumed that students who perceive the educator's dialogic performance as positive are supported in a positive art experience (AE). Thus hypotheses H1, H2, H3, H4 and H5 are stated.

Viewing spectacular motifs in paintings which challenge preconceptions of normality, or experiencing colorful and large art works in halls of art museums, are examples of central elements in art experience. The experience can involve a state of heightened awareness (Pelowski & Akiba 2011, p.81), and it can evoke curiosity and spark the feeling of wanting more information, hence inspiring

reflecting and initiate questions of what, how and why? Art experience (AE) is thus assumed in the model to be indicated by the felt situational interest, attention and feeling “I want to know more”. As art experience (AE) is associated with interest and inherits dispositions for action (Ganger 2006, p.63), it is hypothesized to relate to participation in both the educator dialogue (PED) and the group work discussions (PGWD) and to overall enjoyment of the workshop (ENW). Thus H8, H9 and H10 are added to the model.

The third preconditional factor regards the phase of group work processes. Here students - in a controlled free choice environment - are given a task to solve and in this process discuss the meanings and interpretations of art works. This process offers a framework yielding feelings of engagement (Erstad & Gilje 2008, p.222), an environment of imagination, a degree of feeling free to express personal views (Yardi 2008, p.107), which are assumed to be supportive for discussion and negotiation of content. The proposed construct is therefore labeled “perceived qualities of media production process” (PQMPP). Learning possibilities in this process are linked to thinking and articulating interpretations and meanings, and for students to develop own opinions (Drotner 2008, p.169) when views are exchanged among peers (Sinker 2000, p.208). Because of these qualities and the fact that digital media content in digital photos and audio recordings embrace several modalities (Considine, Horton, & Moorman 2009, p.473), it is hypothesized that perceptions of this environment relate to both participation in group work discussions (PGWD), positive potential for reflection (PPR), and furthermore, that engagement during the process may lead to an evaluation of the workshop as overall enjoyable (ENW). These relations are stated in hypotheses H12, H13 and H15.

Processual factors supporting learning possibilities

As described above, student disinterest, disaffection and a perception of little value in activities, i.e. “amotivation” (Ryan & Deci 2000, p.59) is a strong feeling among some students. This is conceptualized as one of the factors crucial to understanding processes of learning. This construct and the other process factors are incorporated in the conceptual framework as mediators for the relations outlined above. It is assumed that perception and evaluation of the three preconditional factors relate negatively to amotivation (AM), which is depicted in hypotheses H6, H11 and H14. A high degree of positive art experience (AE) is therefore assumed to relate to a low degree of amotivation (AM), and likewise high degrees of perceiving the educator dialogic performance (PEDP) and qualities of the media production process are anticipated to link to low amotivation (AM). The mediating function of amotivation is construed in the following way: If students evaluate the art experience (AE) positively, this is assumed to reduce amotivation, which in turn is argued to both increase levels of participation and the overall enjoyment of the workshop, because students who feel less amotivated will tend to engage and to enjoy the activities. This capacity of amotivation as an inverse mediator, result in three more hypotheses in the conceptual model: H20, H21, and H22. However, amotivation was also shown in the preliminary study of typical viewpoints to be an enduring emotional state, which is likely to result in students leaving the museum with little knowledge or potential for reflecting further gained (Kobbernagel 2013). Consequently, amotivation is at the same time assumed to link to a negative potential for further reflection (NPR). Thus, hypothesis H23 is formulated to designate this possible negative dimension.

A second factor theorized to support learning possibilities is the degree to which students perceive the atmosphere as positive and inclusive with respect to dialogue (PDA). In their reports of art gallery educational activities in five large art museums in Copenhagen, Hjort & Larsen (2003, p.18) and Bøje, Kampmann & Larsen (2005), explain that as some students tend to feel alienated in art intellectual discourse and overtaken by authoritative interpretations put forward by the educator it is important that educators reduce this through supporting students to tell their views and create symmetric dialogic relations. Perceptions of this aspect of the communicative atmosphere are thus anticipated to support learning outcome (PPR), enjoyment in the workshop (ENW) and the level of participation in both educator dialogue (PED) and group work discussions (PGWD). Therefore the hypotheses H16, H17, H18 and H19 are added to the model to represent the positive effects of students’ perceptions of a dialogic atmosphere.

In Helene Illeris’s work on young people and contemporary art, she proposes social interaction in the form of articulating opinions about the experience as one key condition for enhancing learning (Illeris 2005, p.237). In this study participation is chosen to encompass dialogical communicative behavior. Putting words to experience is one of the vehicles for entering processes of accommodating and assimilating knowledge. Dysthe takes this one step further by arguing for the importance of both verbalized and non-verbalized conversation in social interaction and hence opportunity to learn. She

explains that dialogue is not only a facilitator for learning processes in its verbalized form, dialogue also means an inner process of thinking inspired by dialogical qualities (Dysthe 1999, p.40). Dysthe's position is supported in Mustapha & Rahman's (2011) more recent findings which speak of four important ways to participate in learning contexts, among which is passive non-verbal participation; they find clear evidence of how silent behavior can induce attention, involvement and thus possibly entail learning (Mustapha & Rahman 2011, p.155). The meaning of participation in the context of a digital workshop is therefore clarified with both aspects. The constructs proposed to cover participation: participation in group work discussion (PGWD), and participation in educator dialogic (PED) are assumed to link directly to learning potential and through the evaluation of the workshop as a felt, enjoyable event (ENW). The former is argued for in the work of Illeris and Dysthe, and for the latter, Jacky Lumby explains in her work that enjoyment and learning are likely to relate because participation has to do with social interaction among peers, and working with friends in a collaborative process (Lumby 2010, p.256). This adds four hypotheses to the model: H24, H25, H28, and H29.

Evaluation of workshops seen from the perspective of museums' strategic wish to build relationships, also has to incorporate the question of whether students enjoyed the visit (ARKEN 2010). Overall enjoyment of the event thus both secures a basis for receiving students as future guests, and in the context of art learning, enjoyment facilitates a transformation of disinterest to openness towards art, and thus the state of enjoyment is assumed to increase learning. In this context, overall enjoyment of the workshop is indicated by feelings of pleasure, excitement, and that it was fun to take part in the workshop. The construct is thus defined at a more general abstract level than in, for example, the enjoyment of using a website (Lin, Gregor, & Ewing 2008, p.42), and connects more to the cumulative retrospect emotion, as Lumby formulates: "Trait emotions are built over time and refer retrospectively to cumulative experience in enjoying classes. State emotions are experienced in the present, the current enjoyment of a specific class. Emotional traits are therefore predicated on the persistent experience of particular emotional states" (Lumby 2010, p.250). Along these lines it is argued that a student's evaluation of the workshop as enjoyable will likely increase the positive potential for reflection (PPR) and decrease the negative (NPR). Thus, hypotheses H30 and H31 are added to the conceptual model.

Outcome factors

The operationalization of outcome is inspired by, but does not adopt in full, two other constructs: 'Critical Thinking Disposition' (CTD) and Student Approach to Learning (SAL). In regard to the former, Nancy Lampert argues that "…reflecting on multiple interpretations of subject matter is an aspect of critical thinking, so it stands to reason that engagement in critical and aesthetic inquiry fosters in art students a disposition to think critically*"* (Lampert 2006, p.216). Critical thinking is thus preparation for engaging in reflections in relevant situations, however it is not probable that students in a three-hour event achieve this, rather this is seen as a possibility in the present study.

Whereas CTD concerns with more cognitive categories, the concept of SAL deals partly with a total of attitudes and skills, e.g. intentions and motives (Diseth 2007, p.188; Kember, Biggs, & Leung 2004, p.262), defined to capture, what could be termed 'an orientation towards learning', and partly a more substantive aspect which capture the way student's reflect. The SAL instrument contains two subdimensions denoting a 'deep approach to learning' and a 'surface approach to learning'. The first subdimension has a scale, including items covering both student's thinking and behavior, i.e. interest, seeking of meaningfulness, and relating ideas. The second subscale includes items of lack of purpose, and unrelated memorising (Diseth 2007, p.194). The other part relates to educational conditions, but these are not included here, because the outcome factors are intended to tap into what students think they experienced in terms of learning. Outcome factors are thus defined as students' evaluations of learning possibilities, such as: feelings of meaningfulness in the workshop; recognition of the fact that viewing and experiencing art also depends on the viewer positioning; intention to use the content in the future; and a changed view on art.

However, perceptions of meaning and intention are reasonably linked to interest and some degree of affection towards the experience. As some students seem persistently disinterested, it is meaningful to conceive of the outcome in two ways: a negative - which entails a degree of rejection of content; and a positive - entailing a degree of accommodating the content. The case of rejection is also discussed by Illeris, but it is unclear, if this state involves possibilities to learn in some way (Illeris 2005, p.239). Despite simplification, the two constructs, positive and negative potential for further reflection about art (PPR and NPR) provide a way to use the model to evaluate the strength of preconditional and process factor's role as antecedents of learning possibilities. As explicated above, theory argues that

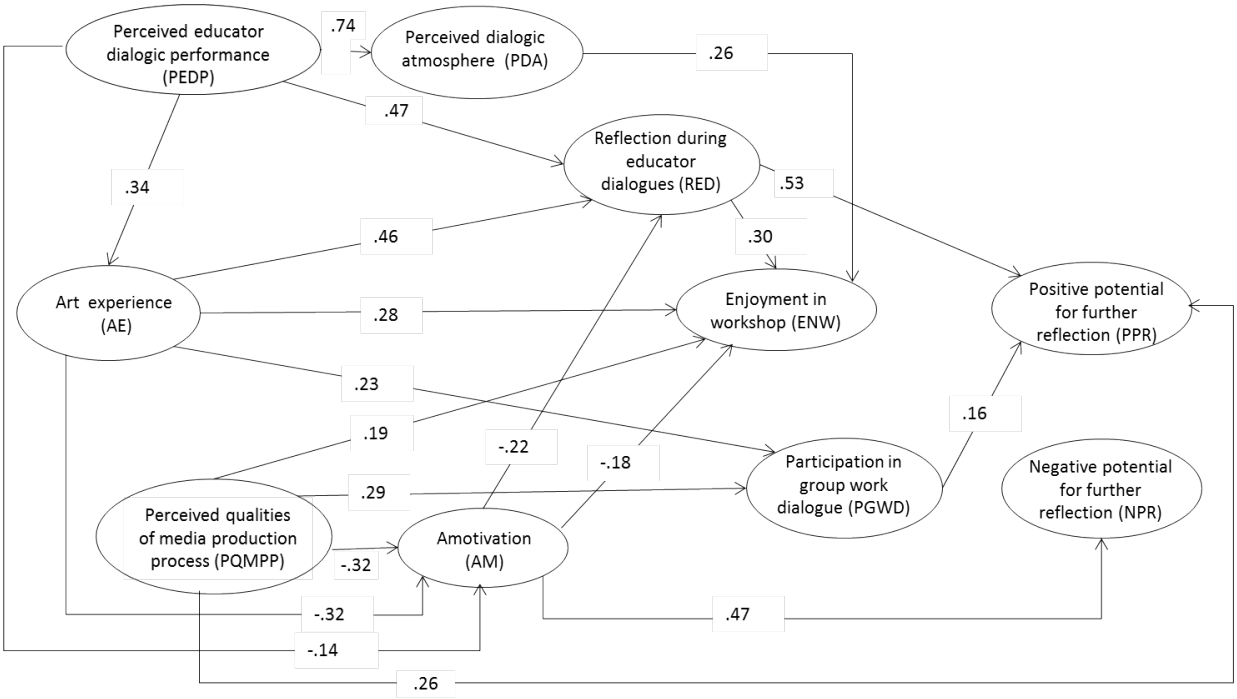
participation is a central process factor for enhancing learning possibilities, hence participation relates to a positive outcome, but a positive experience of participating in dialogue with the educator or in the group work may also decrease the negative outcome factor (NPR). It is probable that the tendency as conceptualized in the model is double-sided, so that, overall experiences transform the perceptions entailing an attitude of little value gained. Lumby shows that students who are active often describe that they achieved something during the event (Lumby 2010, p.258). This is likely also in cases where participation includes verbal and non-verbal dialogue. Consequently two more hypotheses are added H26 and H27 to denote inverse linkage between the participatory constructs and negative potential for reflection, meaning a high level of participation relates to a low level of the negative outcome.

Data collection and analysis

In total 53 indicators were formulated to measure the concepts explicated above and these were written into a questionnaire used for data collection. During autumn 2010 and spring 2011, 670 students attended a digital workshop, and responded on paper sheets to the questionnaire directly after the workshop. In a missing value analysis, systematic non-responding was found in 14 workshops containing 168 student responses, which were removed. The resulting sample contained 502 responses and of these 199 visited ARKEN (96 in "The Media" workshop, and 103 in "The Wild 80s" workshop). 303 visited The National Gallery Denmark (137 attended the "National identity" workshop, and 166 attended the "Portrait, staging and identity" workshop). In total 246 girls and 213 boys completed the survey (Mean of age =15.5, and SD age = 1.68).

Following the procedures in SEM estimation of the model, results are shown as regression weights in figure 3. These are all significant p<.01. The model achieved the fit indexes (recommended values in bracketed parentheses) of X^2 = 1318.5, (p<.0001 df=496)[p>.01 or better >.05], CMIN/df = 2.68 [<.3], CFI = .92 [>.90], TLI = .91 [>.90], SRMR = .13 [<.10] , and RMSEA = .06 [between .05 and .08] (Arbuckle 2010; Brown 2006, p.87).

Figure 3: Final model of factors enhancing possibilities of learning*



**Estimates are standardized (beta) partial regression weights, and all estimates shown are significant at p<.01.*

These values are for the most part above or near the cut-off points, and the model can be argued to fit data with minor caveats. For more about model fit see Paul Barrett's discussion (2007). By examining the beta weights it is possible to initially list interpretations. Interpretation of a partial regression in SEM is based on the translations: .10 is weak, .25 is medium or moderate, and .4 is strong (Schumacker & Lomax 2010, p.128). It should be noted that one of the constructs, 'participation in educator dialogue' (PED) was not confirmed in confirmatory factor analysis. The construct of reflection during educator dialogues proved both more meaningful and statistically reliable. Therefore PED is changed to RED.

Results

1) Dialogic communication seems to be a powerful approach to enhancing students' possibilities of learning. This is especially seen in the relations PEDP->RED beta =.47, p<.01 and RED->PPR beta =.53, p<.01. Students who perceive the educator as respecting opinions, listening and encouraging dialogue about experiences, seem more likely to engage in reflection during dialogues. This increases the likelihood that students acquire a potential for reflecting further after they leave the museum. The dialogic effort of the educator furthermore is likely to support a positive art experience (PEDP->AE beta = .34, p<.01), which in turn seems to increase reflective thinking during the dialogues (AE->RED beta = .46, p<.01) and students' perceptions of a positive dialogic atmosphere.

2) A dialogic atmosphere (PDA) does not seem an important attribute of conditions enhancing possibilities of learning. Perceptions of a good mood, comprehension, and support for sharing personal experiences are not important attributes in the learning perspective, as the only significant relation of perceptions of dialogic atmosphere is PDA->ENW beta = .26, p<.01. This means that dialogic communication as clarified in this model to a lesser degree involves students perceiving the atmosphere; dialogic performance is rather about evoking reflections. Apparently, the model does not support an argument for perceived dialogic atmosphere as a relation important for engaging in reflections during the dialogue with the educator or participating in the group discussions.

3) Art experience, indicated by students' attention and wishes to know more, seems supportive of reflecting during the educator dialogue as noted, and also the discussions in group work processes (AE->PGWD beta = .23, p<.01), which in turn increase the level of positive potential for reflection (PGWD->PPR beta = .16, p<.01). The latter is however remarkably weak and it suggests a difference in strength between the two forms of participation, that is, group work discussions where students talk about their interpretation, discuss meaning with peers, and develop their own opinions are, compared to the educator dialogue, a much weaker factor enhancing learning possibility.

4) The qualities of the process in working with media enhance the possibilities of learning through the supportive and motivating frame for the group work discussions about the art works and the content of the task. This is shown in PQMPP->PGWD beta = .29, p<.01. The qualities of producing with media also seem supportive of students' overall enjoyment of the workshop (PQMPP->ENW beta = .19, p<.01), but participation in discussions and developing opinions about art does not relate to students' overall enjoyment of the workshop. However, the qualities of media production process seem to link moderately to positive potential for reflection as seen in PQMPP->PPR beta = .26, p<.01. Thus, the model suggests that digital media use overall is a strong factor enhancing learning possibilities.

5) Overall enjoyment of the workshop (ENW) does not relate positively to positive potential for reflection nor does it link negatively to negative potential for reflection. That means that despite the fact that the perceptions and behavior of students, as noted in the points above, may involve the emotional state of enjoyment, the overall enjoyment of the workshop does not seem to be a factor enhancing possibilities of learning. Thus increased likelihood of learning is linked to the activity of reflecting during the educator dialogue or in the group work discussions and not to the fact that students see the workshop as enjoyable.

6) Relations of amotivation on the other hand seem to play an important role. Amotivation is likely to be reduced if students evaluate the educator's dialogic performance positively (PEDP->AM beta = -.14, p<.01), and a positive art experience appears in the same (but stronger) to decrease level of amotivation (AE->AM beta = -.32, p<.01). Similarly a positive perception of qualities of working with digital media seem to decrease amotivation (PQMPP->AM beta = -.32, p>.01). Furthermore and importantly, less amotivated students are more likely to listen, see new things in the art works and think further about the themes raised, than highly amotivated students, because amotivation relates moderately negative to reflection during dialogues with the educator (AM->RED beta = -.22, p>.01). These results suggest that of the three preconditional factors, art experience and perception of qualities of the media production process are the strongest conditions to reduce amotivation. If amotivation is low this in turn links to higher levels of reflection during dialogue and higher overall enjoyment of the workshop. The reducing effect on amotivation, however, cannot be read, as at the same time increasing participation in group work discussions, because this relation is insignificant.

7) Amotivation furthermore appears to link very strongly to negative potential for further reflection (AM->NPR beta = .47, p<.01). That is, a negative outcome is very likely to be experienced by students who cannot see what is gained, who think the activities were meaningless and that it was a waste of time, e.g. those highly amotivated. This aspect of the evaluation of the workshops indicate that amotivation

among students has a possible important interpretation. Because the negative outcome is unrelated to, and hence not reduced by, any of the other factors, amotivation appears to explain why students leave the museum with a negative potential for further reflection. A possible interpretation of this pattern may be that the felt amotivation is so strong, that no other central factors have an impact, and when based on disinterest and little perceived value from investment, there are ony reduced possibilities of enhancing learning.

Final comments

The attempt in this study was to produce a generalized picture of central factors important for enhancing learning possibilities in digital workshops. As shown, it is possible to develop a model, which in terms of statistical criteria reaches an acceptable level, and which, with respect to meaning in the relational pattern connects to the discourse of art gallery learning and digital media use presented in theory. It contributes by presenting a holistic map of relations, as they can be found and analytically constructed through theoretical reflection and examination of the empirical material. It makes suggestions about the nature and effect of selected factors in this particular environment in the four workshops in ARKEN and The National Gallery Denmark. The model shows that the dialogic approach, i.e the dialogic performance of the museum educator is an important antecedent of learning potential, because the model suggests that perceptions measured at an aggregate level show a relative strong pattern of effects. Dialogic performance of museum educators is important, but interestingly the effect is not found related to the perception of the atmosphere as dialogic. It suggests that students are not bothered about atmosphere, and results thus contrast with the findings of Hjort & Larsen (2003) and Bøje et al. (2005). Over-intellectualization and the feeling of being an alien in art elite discourse is not necessarily a factor of importance. This resonate well with the characterization of young people as responsible, knowing and decisive about what, when and how to learn about content (Illeris 2005).

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